

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

Year/Sem: III / II

Assignment Number: 9.1 / 24

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Batch: 46

Lab Objectives

- Understand automatic documentation generation
- Write docstrings and comments
- Improve code readability and maintainability
- Compare different documentation styles

Problem 1: find_max Function

Original Code

```
def find_max(numbers):  
    return max(numbers)
```

Docstring

```
def find_max(numbers):  
    """  
    Returns the maximum value from a list of numbers.  
    """  
    return max(numbers)
```

Inline Comments

```
def find_max(numbers):  
    # Find maximum element in list  
    return max(numbers)
```

Google Style

```
def find_max(numbers):  
    """  
    Finds the maximum number.  
  
    Args:  
        numbers (list): List of integers.  
  
    Returns:  
        int: Maximum value.  
    """  
    return max(numbers)
```

Comparison

- Docstrings give function-level explanation
- Inline comments explain specific lines
- Google style is detailed and professional

Problem 2: Login Function

Code

```
def login(user, password, credentials):  
    return credentials.get(user) == password
```

Documentation

```
def login(user, password, credentials):  
    """  
    Verifies user login credentials.  
  
    Args:  
        user (str): Username
```

```

        password (str): Password
        credentials (dict): Stored credentials

Returns:
    bool: True if valid login
"""
return credentials.get(user) == password

```

Problem 3: Calculator Module

```

def add(a, b):
    """Returns sum of two numbers"""
    return a + b

def subtract(a, b):
    """Returns difference"""
    return a - b

def multiply(a, b):
    """Returns product"""
    return a * b

def divide(a, b):
    """Returns quotient"""
    return a / b

```

Problem 4: Conversion Module

```

def decimal_to_binary(n):
    return bin(n)

def binary_to_decimal(b):
    return int(b, 2)

def decimal_to_hexadecimal(n):
    return hex(n)

```

Problem 5: Course Module

```

courses = {}

```

```
def add_course(course_id, name, credits):  
    courses[course_id] = (name, credits)  
  
def remove_course(course_id):  
    courses.pop(course_id, None)  
  
def get_course(course_id):  
    return courses.get(course_id)
```

Conclusion

Documentation improves readability, maintainability, and usability of code.

AI tools can automatically generate high-quality documentation, reducing developer effort.